

Hibernate 2-Week Roadmap (Basic to Advanced)

This roadmap is designed for learners who already know Core Java, OOPs, Collections, JDBC basics, and MySQL basics.

Day 1 — Introduction to ORM & Hibernate

- What is ORM
- Problems with JDBC
- What is Hibernate
- Hibernate Architecture
- JPA vs Hibernate
- Advantages of Hibernate
- Create Maven project
- Add Hibernate dependencies
- Connect MySQL with Hibernate
- Create simple Student entity

Day 2 — Hibernate Configuration

- hibernate.cfg.xml
- Dialects
- Driver class
- Database URL, username, password
- SessionFactory
- Session
- Transaction
- Insert one record into DB

Day 3 — Entity Mapping Basics

- @Entity
- @Table
- @Id
- @GeneratedValue
- @Column
- Primary keys
- Employee CRUD table

- Student Management mini project

Day 4 — CRUD Operations

- save()
- persist()
- get()
- load()
- update()
- merge()
- delete()
- Difference between get() vs load()
- Difference between update() vs merge()

Day 5 — HQL & JPQL

- What is HQL
- Select queries
- Named parameters
- Pagination
- Aggregate functions
- Fetch top employees
- Count records

Day 6 — Criteria API & Native SQL

- Criteria API basics
- Restrictions
- Native queries
- SQLQuery / NativeQuery
- Dynamic search filters

Day 7 — Revision + Project

- Employee Management Project
- Add employee
- Update employee
- Delete employee

- Search employee
- HQL queries revision

Day 8 — Relationship Mapping

- @OneToOne
- @OneToMany
- @ManyToOne
- @ManyToMany
- mappedBy
- Owning side
- Student-Course project
- Customer-Orders project

Day 9 — Fetching & Cascading

- Lazy loading
- Eager loading
- Cascade types
- Orphan removal
- Parent-child operations

Day 10 — Inheritance Mapping

- Single Table strategy
- Joined strategy
- Table per class
- @Inheritance
- @DiscriminatorColumn
- Vehicle-Car-Bike example

Day 11 — Component Mapping & Embeddables

- @Embeddable
- @Embedded
- Value objects
- Address inside Employee entity

Day 12 — Caching in Hibernate

- First-level cache
- Second-level cache
- Query cache
- Ehcache basics
- Observe SQL reduction

Day 13 — Advanced Concepts

- Hibernate lifecycle states
- Transient state
- Persistent state
- Detached state
- Removed state
- Dirty checking
- Interceptors
- Event listeners
- Batch processing
- Bulk insert operations

Day 14 — Final Project + Spring Data JPA Transition

- Library Management System
- Book CRUD
- Author mapping
- Issue/Return books
- Search operations
- Pagination
- Lazy loading
- Transition to Spring Data JPA

Recommended Next Learning Path

- Spring Framework basics
- Spring Boot
- Spring Data JPA
- REST APIs
- Security

- Microservices

Daily Practice Recommendation

- 2 hours theory
- 3 hours coding
- 1 hour debugging
- 30 minutes revision